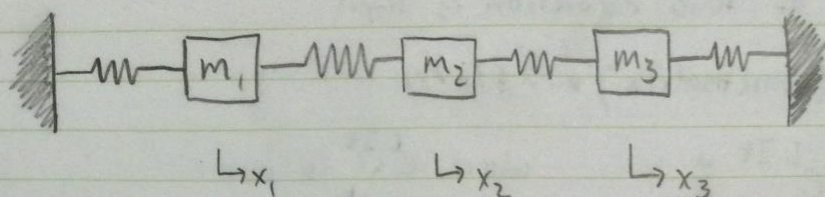


Question 42

Andrew Tan act63



★ unforced and undamped $\Rightarrow$ homogenous solution

★ masses m_1 m_2 m_3

★ All equal springs with equal rest lengths

GOAL: find some form of $\vec{x}(t) = A \vec{x}_0$

mass matrix $M = \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}$

$\vec{x}_0 = IC = \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ \dot{x}_{1,0} \\ \dot{x}_{2,0} \\ \dot{x}_{3,0} \end{bmatrix}$

stiffness matrix $K = \begin{bmatrix} 2k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & 2k \end{bmatrix}$

using IC, M , K , we want to find the positions of all 3 masses...

$\vec{x} = \vec{x}_0 \times \underbrace{\text{(some matrix)}}_{\text{is the matrix exponential of}}$

this is a 3-by-3 zero Matrix because our system has 0 damping

$M^{-1}K = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

this is a 3-by-3 Identity Matrix because our system has 3 DOF.

so a detailed version of this equation is then

$$\vec{x} = (\text{some matrix}) \times \vec{x}_0$$

$$\vec{x} = e^{[J]t} \vec{x}_0$$

where $e^{[J]t}$ is our matrix exponential times time t .

$$\vec{x}(t) = e^{\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \boxed{M^{-1}K} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} t} \vec{x}_0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \expm \left(\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \boxed{M \backslash K} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} t \right) \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ \dot{x}_{1,0} \\ \dot{x}_{2,0} \\ \dot{x}_{3,0} \end{bmatrix}$$

→ this is the motion of our system given any initial conditions

Contents

- [system paramters](#)
- [get x\(t\) from x0](#)
- [plot](#)
- [animate](#)

```
% MAE 5730 Q42 - Andrew Tan aet63

% Q42 solution - For the three mass system
% find the motion if given IC for pos and vel.
% In some normal modes, check the motion
% against direct integration of the ODEs.
% 3 masses, 4 equal springs
close all; clear all; clc;
```

system paramters

```
m1 = 1; m2 = 1; m3 = 1;
k = 1;
klength = 2;           %spring rest length
t = linspace(0,20,400);

M = [m1 0 0 ;          %mass matrix
     0 m2 0 ;
     0 0 m3];
K = [-2*k k 0 ;        %stiffness matrix
     k -2*k k ;
     0 k -2*k];
A= [zeros(3,3) eye(3) ;
    M\K        zeros(3,3)];

%IC= [x1 x2 x3 x1d x2d x3d]
x0 = [rand rand rand 0 -rand 0 ]'; % measured from each mass's equilibrium
```

get x(t) from x0

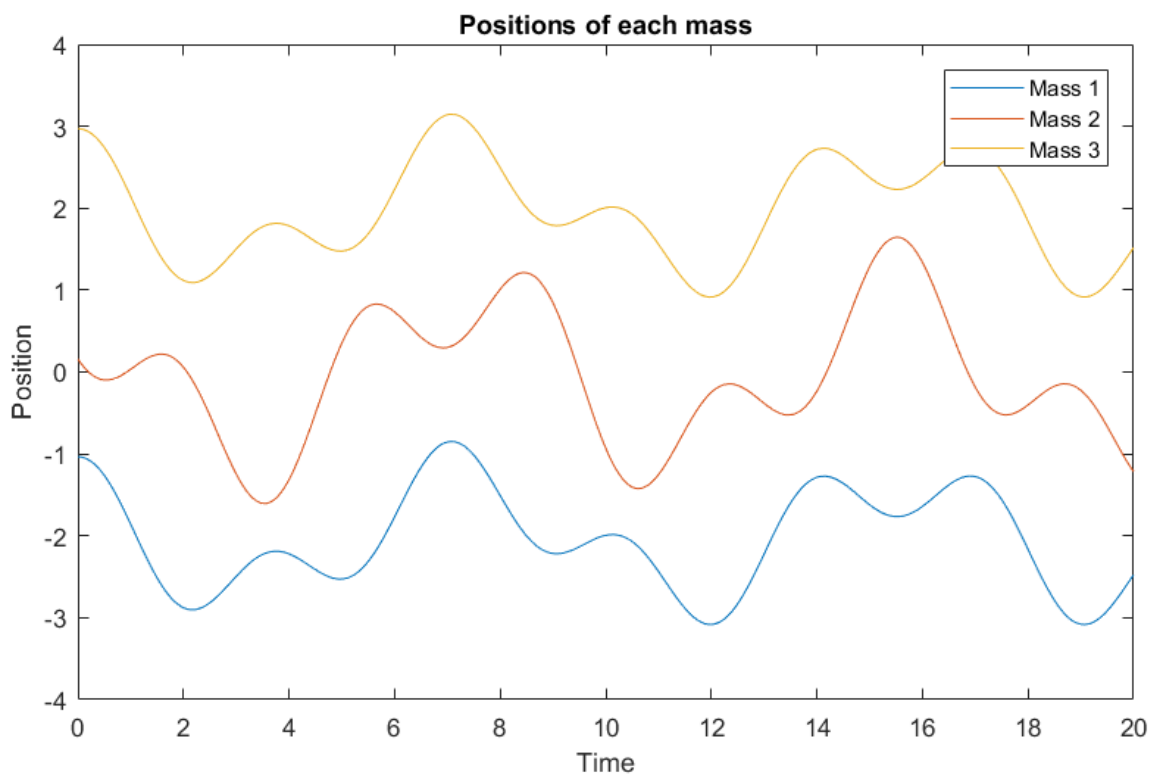
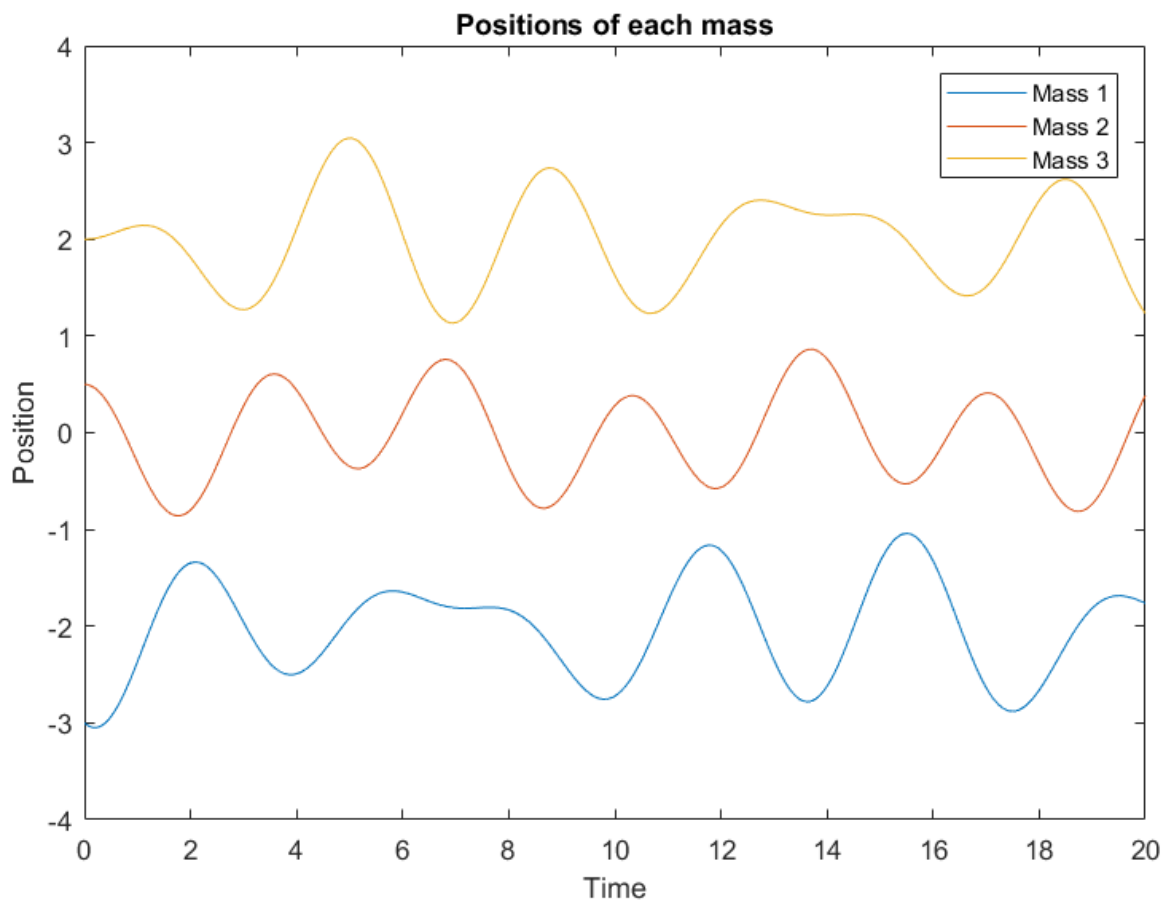
```
x = zeros(6,length(t)); %initialize

for i = 1:length(t)
    x(:,i) = expm(A*t(i))*x0; %each column describes the system at time t
end
```

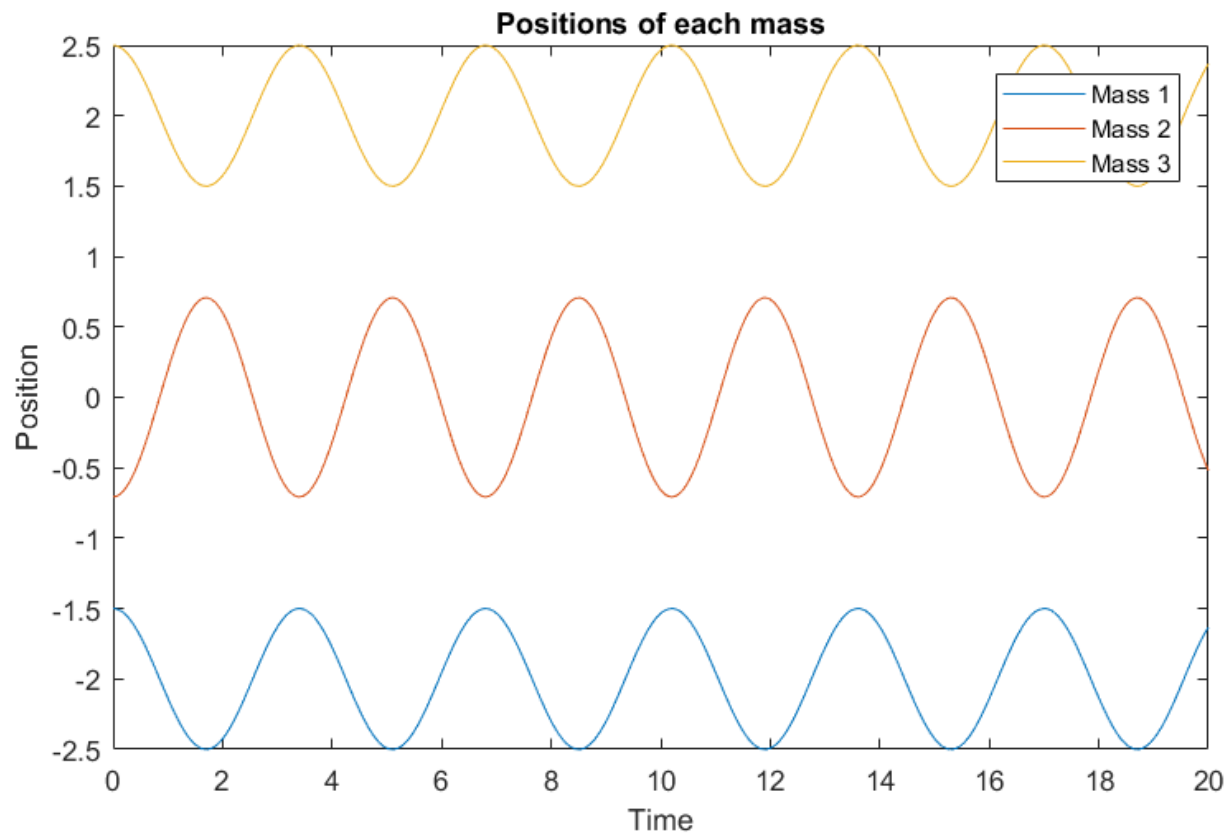
plot

```
x_p = x + [-klength 0 klength 0 0 0]'; %positions measured from origin

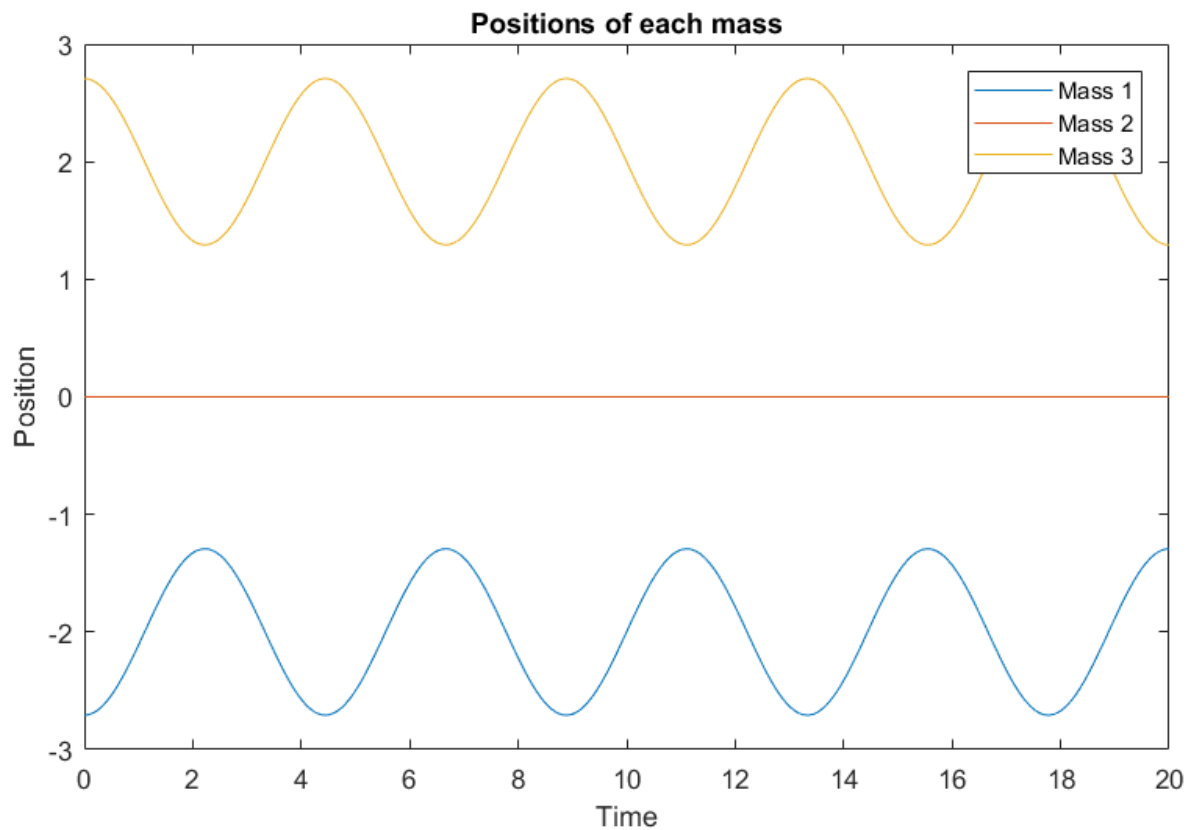
figure('units','normalized','outerposition',[0 0 1 1]) %fullscreen
plot(t,x_p(1:3,:))
title('Positions of each mass')
legend('Mass 1','Mass 2', 'Mass 3')
xlabel('Time'); ylabel('Position')
```

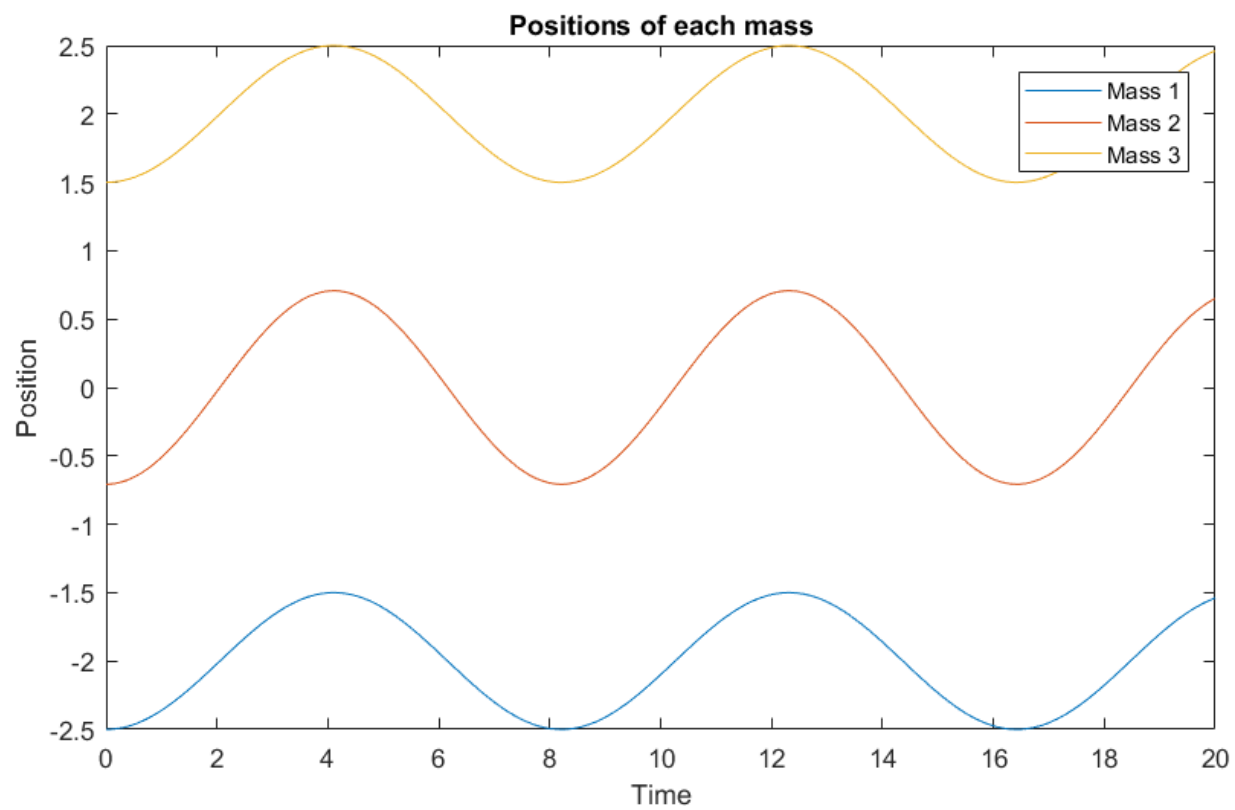
*Both figures above are generated with random initial conditions.



One of three normal modes: $x_1 = 0.5, x_2 = -0.707, x_3 = -0.5, \dot{x}_1 = \dot{x}_2 = \dot{x}_3 = 0$



Another of three normal modes: $x_1 = 0.707, x_2 = 0, x_3 = -0.707, \dot{x}_1 = \dot{x}_2 = \dot{x}_3 = 0$



Last of three normal modes: $x_1 = -0.5, x_2 = -0.707, x_3 = -0.5, \dot{x}_1 = \dot{x}_2 = \dot{x}_3 = 0$